THE COLLISION INVARIANT

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Abstract. For a prime p and base b, the digit function $\delta(r) = \lfloor br/p \rfloor$ partitions the residues $\{1, \ldots, p-1\}$ into b contiguous bins. The *collision count* $C(g)$ records how many residues share a bin with their image under multiplication by g. We prove four results about this function.

First, the *gate width theorem*: exactly $b-1$ multipliers satisfy $C(g) = 0$, given by the explicit family $g = -u/(b-u) \bmod p$ for $u = 1, \ldots, b-1$. The proof rests on a linearization that transforms the contiguous Beatty bins into residue classes modulo b.

Second, the *finite determination theorem*: the collision deviation $S_\ell(p) = C(b^\ell \bmod p) - \lfloor (p-1)/b \rfloor$ depends only on $p \bmod b^{\ell+1}$. This reduces the study of the collision invariant from an analytic problem on primes to a combinatorial problem on a finite group.

Third, the *reflection identity*: $S_\ell(a) + S_\ell(m-a) = -1$ for $m = b^{\ell+1}$, implying a grand mean of $-1/2$ and a pairing symmetry across the group of units.

Fourth, the *half-group theorem*: for every non-trivial diagonal slice n, the wrapping set $W_n = \{a : (n+1)a \bmod m < a\}$ has size exactly $\phi(m)/2$. The bilateral symmetry $a \mapsto m-a$ swaps wrapping with non-wrapping.

1. Introduction

For a prime $p > b$, the digit function $\delta(r) = \lfloor br/p \rfloor$ assigns to each nonzero residue modulo p its leading digit in base b. It partitions the residues into b bins $B_d = \{r : \delta(r) = d\}$, each a contiguous interval of length $\lfloor (p-1)/b \rfloor$ or $\lfloor (p-1)/b \rfloor + 1$. This partition is a Beatty partition [4] determined by the three-distance theorem applied to the sequence $\{r/p\}$ at scale $1/b$.

Digit functions of this type arise in the study of digit sums of primes [2], character sums over digit-constrained sets [3], and the statistical properties of decimal expansions [1]. The present paper studies a specific invariant derived from the interaction between the bin partition and the multiplicative structure of $(\mathbb{Z}/p\mathbb{Z})^\times$.

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Definition 1. For $g \in (\mathbb{Z}/p\mathbb{Z})^\times$, the *collision count* is

$$C(g) = \#\{r \in \{1, \ldots, p-1\} : \delta(r) = \delta(gr \bmod p)\}.$$

The collision count measures how many residues share a bin with their image under multiplication by g. It is a function on the multiplicative group that encodes the interaction between the additive structure of the bins and the multiplicative action of g.

The *collision deviation* at lag ℓ is

$$S_\ell(p) = C(b^\ell \bmod p) - \left\lfloor \frac{p-1}{b} \right\rfloor,$$

the departure of the collision count at the specific multiplier $g = b^\ell$ from the bin size Q.

2. The Linearization

The key structural observation is that multiplication by b transforms the bin partition into a congruence partition.

Lemma 2 (Conjugation)**.** *For $r \in \{1, \ldots, p-1\}$, let $[x]_p = x \bmod p$ and set $x = [br]_p$. Then $br = p\,\delta(r) + x$ and*

$$\delta(r) = \delta(s) \quad \Longleftrightarrow \quad [br]_p \equiv [bs]_p \pmod{b}.$$

Under the permutation $r \mapsto [br]_p$, the bins B_d are exactly the residue classes modulo b.

Proof. Since $br = p\lfloor br/p \rfloor + (br \bmod p) = p\,\delta(r) + x$, reducing modulo b gives $0 \equiv p\,\delta(r) + x \pmod{b}$. Since $\gcd(p, b) = 1$, the class of x modulo b determines $\delta(r)$ uniquely. □

This linearization converts the geometric problem (which residues lie in the same contiguous interval?) into an algebraic problem (which residues are congruent modulo b?). It is the foundation of all subsequent results.

Lemma 3. *For every $g \in (\mathbb{Z}/p\mathbb{Z})^\times$,*

$$C(g) = \#\{x \in \{1, \ldots, p-1\} : x \equiv [gx]_p \pmod{b}\}.$$

Proof. Apply Lemma 2 to the pair r and $gr \bmod p$. □

3. The Gate Width Theorem

Lemma 4. *For $g \neq 1$, define $c \equiv b(1-g)^{-1} \pmod{p}$ with $1 \leq c \leq p-1$. If $1 \leq c \leq b-1$, then $C(g) = 0$.*

Proof. A collision requires $y = gx \bmod p$ with $y \equiv x \pmod b$, so $y = x + mb$ for some integer m, and $x \equiv -mc \pmod p$.

If $m \geq 0$: from $x + mb \leq p-1$ and $mc < p$ (since $c \leq b-1$ and $m \leq Q$), we get $1 \leq x + mc \leq p-1-m(b-c) \leq p-1$, so $x + mc$ is a positive integer strictly less than p, contradicting $x + mc \equiv 0 \pmod p$.

If $m < 0$: write $m = -n$ with $n \geq 1$. From $x - nb \geq 1$, we get $0 < x - nc = (x-nb) + n(b-c) \leq p-1$, contradicting $x - nc \equiv 0 \pmod p$. □

Lemma 5. *If $c \geq b + 1$, then $C(g) \geq 1$.*

Proof. Take $x = p - c$. Then $1 \leq x \leq p-b-1$, so $y = x + b$ lies in $\{1, \ldots, p-1\}$ with $y \equiv x \pmod b$. Since $c(1-g) \equiv b$, we get $gx \equiv x + b = y \pmod p$, producing a collision. □

Theorem 6 (Gate width)**.** *For any prime $p > b$,*

$$\{g \in (\mathbb{Z}/p\mathbb{Z})^\times : C(g) = 0\} = \left\{ -\frac{u}{b-u} \bmod p : u = 1, \ldots, b-1 \right\}.$$

In particular, exactly $b - 1$ multipliers have $C(g) = 0$, independent of p.

Proof. For $g = 1$: $C(1) = p - 1 > 0$. For $g \neq 1$: $c = b(1-g)^{-1}$ is well defined. The case $c = b$ would give $(1-g)^{-1} \equiv 1$, hence $g \equiv 0$, impossible for a unit. So $c \in \{1, \ldots, b-1\} \cup \{b+1, \ldots, p-1\}$. By Lemmas 4 and 5, $C(g) = 0$ if and only if $c \in \{1, \ldots, b-1\}$. Setting $u = b - c$ gives $g = -u/(b-u) \bmod p$ for $u = 1, \ldots, b-1$. These are distinct since $p > b$. □

Remark 7. The deranging multipliers form a rational family parameterized by u. The count $b - 1$ depends only on the base, not on the prime. No primitive-root hypothesis is needed.

4. The Finite Determination Theorem

Theorem 8 (Finite determination)**.** *For any base $b \geq 2$, lag $\ell \geq 1$, and integer $p > b^{\ell+1}$ with $\gcd(p, b) = 1$, the collision deviation $S_\ell(p)$ depends only on $p \bmod b^{\ell+1}$.*

Proof. Set $m = b^{\ell+1}$. For $r \in \{1, \ldots, p-1\}$, define the slice index $n(r) = \lfloor mr/p \rfloor$. Since the slice width $p/m > 1$, each slice contains integers. The digit function and the collision multiplier are constant on each slice:

$$\delta(r) = \left\lfloor \frac{n}{b^\ell} \right\rfloor.$$

For the second digit: write $b^\ell r = qp+y$ with $0 \leq y < p$, so $y = b^\ell r \bmod p$. Then $mr = b \cdot b^\ell r = bqp + by$, hence $n = \lfloor mr/p \rfloor = bq + \delta(y)$, giving

$$\delta(b^\ell r \bmod p) = n \bmod b.$$

A slice n contributes to the collision count if and only if $\lfloor n/b^\ell \rfloor = n \bmod b$. Define the diagonal set

$$G_{\ell,b} = \{n \in \{0, \ldots, m{-}1\} : \lfloor n/b^\ell \rfloor = n \bmod b\}.$$

Then $|G_{\ell,b}| = b^\ell$ and

$$C(b^\ell \bmod p) = -1 + \sum_{n \in G} \left(\left\lfloor \frac{(n{+}1)p}{m} \right\rfloor - \left\lfloor \frac{np}{m} \right\rfloor \right).$$

The -1 accounts for the slice $n = 0$ containing $r = 0$, which is excluded from the collision count.

Writing $p = mt + a$ with $1 \le a < m$, each summand becomes $t + \lfloor (n{+}1)a/m \rfloor - \lfloor na/m \rfloor$, and since $Q = b^\ell t + \lfloor a/b \rfloor$:

$$S_\ell(p) = -1 - \left\lfloor \frac{a}{b} \right\rfloor + \sum_{n \in G} \left(\left\lfloor \frac{(n{+}1)a}{m} \right\rfloor - \left\lfloor \frac{na}{m} \right\rfloor \right),$$

which depends only on $a = p \bmod m$. □

Definition 9. For $m = b^{\ell+1}$ and $a \in (\mathbb{Z}/m\mathbb{Z})^*$, define $S_\ell(a)$ to be the common value of $S_\ell(p)$ for any integer $p > m$ with $p \equiv a \pmod{m}$ and $\gcd(p, b) = 1$. Equivalently:

$$S_\ell(a) = -1 - \left\lfloor \frac{a}{b} \right\rfloor + \sum_{n \in G} \left(\left\lfloor \frac{(n{+}1)a}{m} \right\rfloor - \left\lfloor \frac{na}{m} \right\rfloor \right).$$

Remark 10. Primality is not required. The result holds for all $p > m$ with $\gcd(p, b) = 1$. The modulus $b^{\ell+1}$ is sharp: reduction modulo b^ℓ does not determine S_ℓ.

5. The Reflection Identity

Theorem 11 (Reflection)**.** *For $m = b^{\ell+1}$ and any unit $a \in (\mathbb{Z}/m\mathbb{Z})^\times$:*

$$S_\ell(a) + S_\ell(m - a) = -1.$$

Proof. Write $d_n(a) = \lfloor (n{+}1)a/m \rfloor - \lfloor na/m \rfloor \in \{0, 1\}$ for the slice contribution. By Theorem 8:

$$S(a) = -1 - \left\lfloor \frac{a}{b} \right\rfloor + \sum_{n \in G} d_n(a).$$

Interior slices ($1 \le n \le m{-}2$, so $m \nmid na$ and $m \nmid (n{+}1)a$): the identity $\lfloor n(m{-}a)/m \rfloor = n - 1 - \lfloor na/m \rfloor$ gives $d_n(m{-}a) = 1 - d_n(a)$.

Endpoint slices: $d_0(a) = 0$ for all units (since $0 < a < m$), and $d_{m-1}(a) = 1$ for all units (since $\lfloor (m{-}1)a/m \rfloor = a - 1$). Both $n = 0$ and $n = m{-}1$ lie in G, so they contribute $d_0(a) + d_0(m{-}a) = 0$ and $d_{m-1}(a) + d_{m-1}(m{-}a) = 2$.

Since $\lfloor a/b \rfloor + \lfloor (m-a)/b \rfloor = b^\ell - 1$ (because $b \nmid a$ for units), the sum over all $|G| = b^\ell$ diagonal slices gives:

$$S(a) + S(m-a) = -2 - (b^\ell - 1) + 0 + 2 + (b^\ell - 2) = -1. \qquad \square$$

Corollary 12 (Grand mean)**.** *The average of S_ℓ over all units modulo $b^{\ell+1}$ is $-1/2$.*

Proof. The map $a \mapsto m-a$ is an involution on the units. By Theorem 11, each pair averages to $-1/2$. $\square$

6. The Half-Group Structure

The finite determination formula expresses $S_\ell(a)$ as a sum over diagonal slices $n \in G_{\ell,b}$. Each summand $\lfloor (n+1)a/m \rfloor - \lfloor na/m \rfloor$ equals 1 when $(n+1)a$ wraps past a multiple of m and 0 otherwise. The *wrapping set* for diagonal slice n is

$$W_n = \{a \in (\mathbb{Z}/m\mathbb{Z})^\times : (n+1)a \bmod m < a\}.$$

Theorem 13 (Half-group)**.** *For every non-trivial diagonal slice n (with $(n+1) \not\equiv 0, 1 \pmod m$): $|W_n| = \phi(m)/2$. Every diagonal slice except $n = 0$ and $n = m-1$ is non-trivial, since $1 < n+1 < m$ for $n \in \{1, \ldots, m-2\}$.*

Proof. Let $c = n+1$ and $a \in W_n$, so $ca \bmod m < a$. Then $c(m-a) \bmod m = m - (ca \bmod m) > m - a$, so $m - a \notin W_n$.

Conversely, if $a \notin W_n$, then $ca \bmod m \geq a$. Equality is excluded: $ca \equiv a \pmod m$ would require $c \equiv 1 \pmod m$, contradicting the hypothesis. So $ca \bmod m > a$, giving $c(m-a) \bmod m = m - (ca \bmod m) < m - a$, so $m - a \in W_n$.

Thus $a \mapsto m - a$ is a bijection between W_n and its complement, and $|W_n| = \phi(m)/2$. $\square$

The wrapping set is always exactly half the group because the reflection $a \mapsto m - a$ swaps wrapping with non-wrapping. This is the same symmetry that produces the reflection identity: the bilateral pairing of residues governs both the global structure of S_ℓ and the local structure of each diagonal slice.

7. Computational Verification

All results are verified computationally using the `nfield` engine [7].

b	p	Q	deranging count	formula
10	17	1	9	$b-1$
10	97	9	9	$b-1$
10	193	19	9	$b-1$
7	41	5	6	$b-1$
12	67	5	11	$b-1$

TABLE 1. Gate width verified across bases and primes.

b	modulus	classes	determined
3	9	6	yes
5	25	20	yes
7	49	42	yes
10	100	40	yes

TABLE 2. Finite determination verified: S_1 constant within each class modulo b^2.

8. Remarks

The collision invariant S_ℓ is a new arithmetic function on primes arising from the interaction between additive structure (contiguous bins) and multiplicative structure (the action of b^ℓ). The finite determination theorem reduces its study to a combinatorial problem on the finite group $(\mathbb{Z}/b^{\ell+1}\mathbb{Z})^\times$.

The reflection identity provides exact structural information: the grand mean is $-1/2$ and the values pair under $a \mapsto m-a$. These properties are consequences of the bin geometry, not of the distribution of primes.

A companion paper [6] analyzes the prime-sum behavior of S_ℓ via its Dirichlet character decomposition over the finite group, showing that the centered prime harmonic sum converges.

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